

F TALENT CODE 2026

Fair Division of the Ancient Relics

Time Limit: 2 seconds | Memory Limit: 512 MB



Problem statement

In the royal treasury of an ancient empire there exist exactly n sacred relics, numbered from 1 to n . The magical power of relic i is precisely equal to the integer i .

The High Priests must partition all relics into two non-empty groups such that the total magical power of the two groups is identical. Only then can the two protective shields of the empire be charged with perfectly balanced energy.

Because the order of relics inside a group does not matter and swapping the two groups produces the same partition, two partitions that differ only by swapping the groups are considered identical.

Your task is to count the number of distinct ways to perform such a balanced partition. Since the answer may be enormous, output it modulo $10^9 + 7$.

For example, if $n = 7$ there are four solutions:

- $\{1, 3, 4, 6\}$ and $\{2, 5, 7\}$
- $\{1, 2, 5, 6\}$ and $\{3, 4, 7\}$
- $\{1, 2, 4, 7\}$ and $\{3, 5, 6\}$
- $\{1, 6, 7\}$ and $\{2, 3, 4, 5\}$

Input / Output specification

INPUT	OUTPUT
The only input line contains an integer n	Print the answer modulo $10^9 + 7$
	If n is an invalid value, return 0

Constraints

- $1 \leq n \leq 1000$.

Example 1:

INPUT	OUTPUT
7	4

Explanation:

There are exactly **four distinct partitions** of $\{1, 2, 3, 4, 5, 6, 7\}$ into two sets with equal sum 14:

- $\{1, 3, 4, 6\}$ and $\{2, 5, 7\}$
- $\{1, 2, 5, 6\}$ and $\{3, 4, 7\}$

- {1, 2, 4, 7} and {3, 5, 6}
- {1, 6, 7} and {2, 3, 4, 5}

Example 2:

INPUT	OUTPUT
3	1

Explanation:

The only valid partition is {3} and {1, 2}. Both sets sum to 3. Note that the empty set is not allowed, and swapping the two sets does not create a new partition.

Hint: Think about the total sum of numbers from 1 to n. For the two sets to have equal sums, what must be true about this total? If the condition is satisfied, try using dynamic programming to count how many subsets have a sum equal to half of the total, and how to overcome limitation of int/ long long data type when n=1000.